

AI Internship Learning Plan

GitHub= <https://github.com/abeerzaff/AI-Internship-PITB>

My name is Abeer Zafar and I recently graduated in Data Science from PUCIT. Throughout in my academic journey, I have worked on projects involving Python programming, Power BI dashboards, SQL and Django. I enjoy to solve real world problems through technology and continuously strive to improve my skills. As part of my Final year project, I developed Sherlock Droid an AI-powered Android security analysis framework that combines Static and Dynamic analysis with LLM(Large Language Model) models to identify vulnerabilities in Android Applications. I am particularly interested in learning how AI systems work, how they are designed, trained and deployed in real world environment. Through this internship my aim is to gain deeper understanding of emerging AI-technologies, gain hands-on industry experience and further strengthen my skills in AI technologies that will help me to build a successful career in Artificial Intelligence and machine learning.

Current Skillset

- Python Programming
- SQL and Database Management
- Data Visualization and Analysis
- Power BI dashboard
- Django and Django REST API
- GitHub basics
- AI/ML fundamentals

Areas for Improvement

- Prompt Engineering
- Agentic AI and Generative AI systems
- Advance Machine learning and Deep learning
- AI deployment
- Vector databases

Learning Goals

- Learn advance Prompt Engineering techniques
- Understand the architecture , training and work flow of large language models
- Explore Agentic AI systems
- Gain practical experience in building Generative AI applications
- Understands how AI agents works in real world
- Strengthen skills in AI-powered tools
- Strengthen project management skills
- Improve communication and documentation skills

Understanding of AI

Artificial Intelligence is the ability of computers to perform task that normally require human intelligence . Human intelligence include thinking , understanding , making decision and problem solving . AI enables machines and software to perform these types of tasks. Traditional software works using fixed rules to perform tasks but the problem like detecting frauds , understanding , pattern these are complex problems that cannot be solve efficiently using rule based systems . AI is use to make enough machines intelligent to solve these types of problem. AI have so many domains like Natural Language Processing for understanding and processing human language , while Computer vision for understanding images and videos . AI also have important subfields like Machine learning where system learns patterns from data to make decisions or predictions , Deep learning is subdomain of ML that uses neural network to learn deep and complex patterns from large amount of data .

Expectations from Internship

- Learn from experts
- Gain practical project experience
- Improve problem solving skills
- Improve technical skills in prompt engineering
- Learn how to build scalable , secure, and reliable AI powered applications
- Develop industry -ready skills to support my long term career goals in AI
- Improve my technical communication , documentation and presentation skills